

EDA assignment 2-3

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1 Introduction

Earthquakes are among the Earth's most destructive natural hazards and occur as a result of tectonic plate movement along plate boundaries around the world (Lay & Wallace, 1995). Understanding where and when earthquakes occur is an important aspect of seismology, as it contributes to seismic risk and hazard assessment (Gutenberg & Richter, 1956).

This report analyse a combined dataset of recorded earthquakes that span from 2 January 1965 to 17 March 2023. The dataset was created by merging two data sources, a Kaggle earthquake dataset covering the period from 1965 to 2016, and a United States Geological Survey query dataset covering December 2016 to March 2023. Following data cleaning and preprocessing, the report investigates the distributions of key variables, temporal trends in earthquake occurrence, and the spatial distribution of earthquakes across selected regions.

2 Data Wrangling

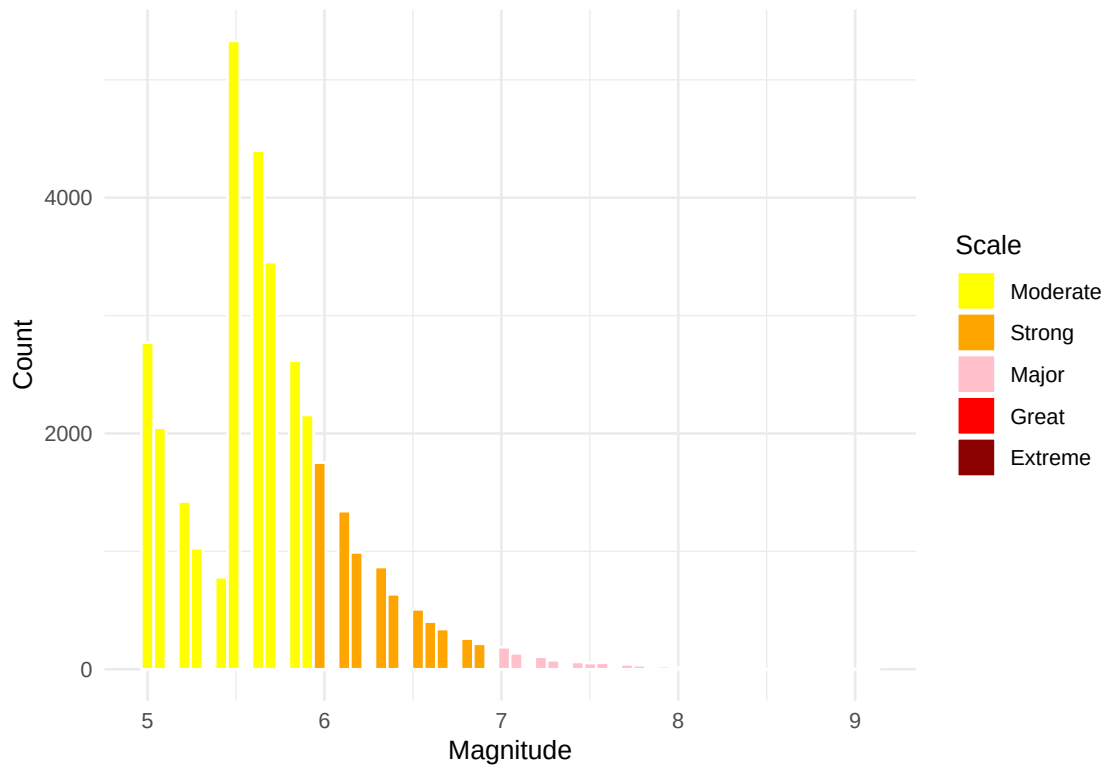
The two raw datasets shared several common variables, although differences were present in naming conventions and date-time formatting. A summary of the changes(common names) to the datasets can be seen in table 1:

Earthquake (1965–2016)	query (2016–2023)	Common name
Date + Time (separate columns)	time	datetime
Latitude / Longitude	latitude / longitude	latitude / longitude
Depth / Depth Error	depth / depthError	depth / depth_error
Magnitude / Magnitude Error	mag / magError	magnitude / magnitude_error
Magnitude Type	magType	magnitude_type
Azimuthal Gap	gap	azimuthal_gap
Horizontal Error	horizontalError	horizontal_error
Root Mean Square	rms	rms
ID / Status	id / status	id / status
Location Source / Magnitude Source	locationSource / magSource	location_source / magnitude_source
Type	type	type

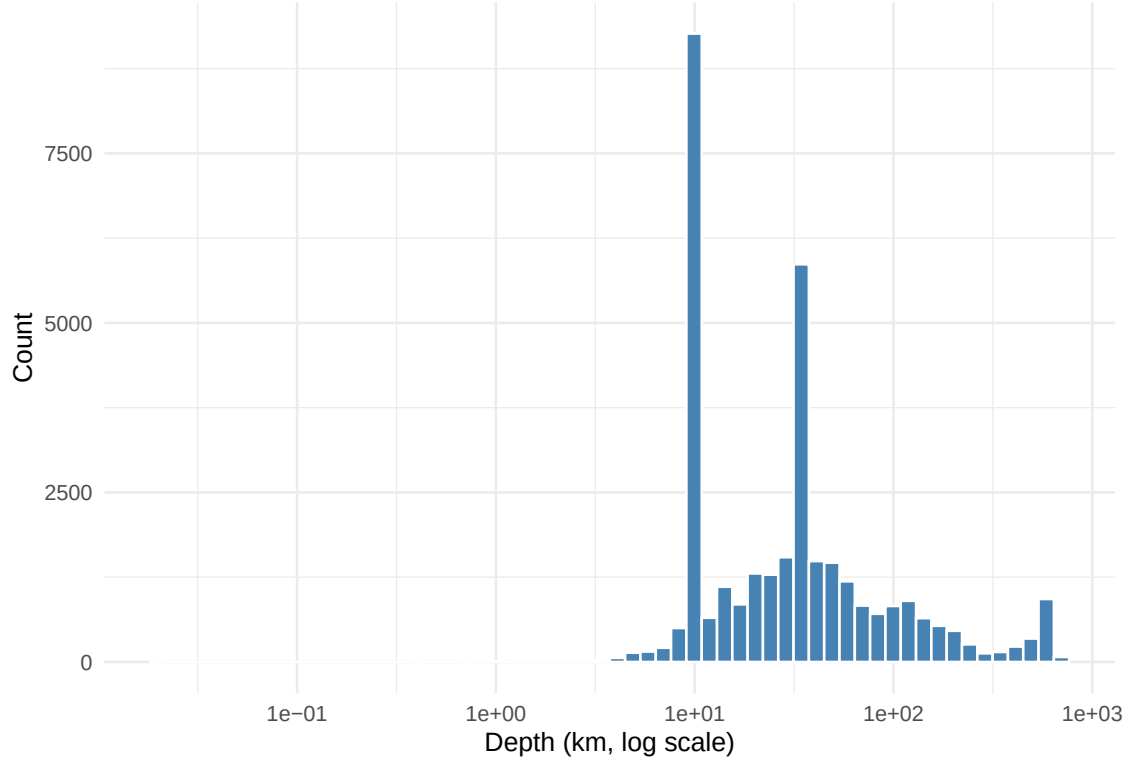
Certain variables were present in only one of the two datasets. These included variables such as “place”, “nst”, and “net” from the query dataset, as well as “Depth Seismic Stations” and “Magnitude Seismic Stations” from the Earthquake dataset. These variables were excluded before merging, as retaining non-common variables would introduce structural issues such as missingness and reduce comparability across observations. After merging and preprocessing the cleaned datasets, the combined dataset contained 34 404 observations with 16 variables of the earthquake records that span from 1965 to 2023. Several variables that can be described as associated with uncertainty showed to have measurements that contained a large proportion of missing values. This could be due to structural differences between the two datasets, as they were recorded at different years, so many older earthquake records probably did not include standardised uncertainty reporting. Variables that showed a lot of these missing values include “depth_error”, “magnitude_error”, “horizontal_error”, and “azimuthal_gap”. These variables were not imputed, as they could potentially cause bias the to analysis. The core variables used throughout the analysis, including “earthquake magnitude”, “depth”, “location”, and “datetime”, contained no missing values after correcting the unstandardised datetime entries. As a result, no observations were removed from the dataset. The dataset contained several seismic event categories, although earthquakes represented the majority of observations. To ensure consistency throughout the analysis, the dataset was filtered to retain only observations classified as “Earthquake”. This brought the number of observations down to 34 168. Then the dataset was followed with a magnitude Scale classification variable following data filtering, earthquake magnitudes were categorised according to the Richter magnitude scale, ranging from Micro to Extreme events. This classification allowed for easier interpretation of earthquake severity throughout the analysis. The distribution of earthquake magnitudes indicated that the dataset was dominated by lower to moderate magnitude events, while extreme earthquakes occurred relatively less often. With the data wrangling done, the next step was to explore the initial patterns and distribution within the dataset.

3 Initial exploration

3.1 feature distribution



The distribution of earthquake magnitudes shows a clear decline in frequency as the magnitude increases. Smaller earthquakes far can bee seen to occur more frequently than large-magnitude events, this reflects the expected global seismic behaviour as we see across the world that extreme earthquakes are rarely reported. Low-magnitude earthquakes were also observed and this can likely reflect detection limitations in earlier decades of earthquake monitoring.

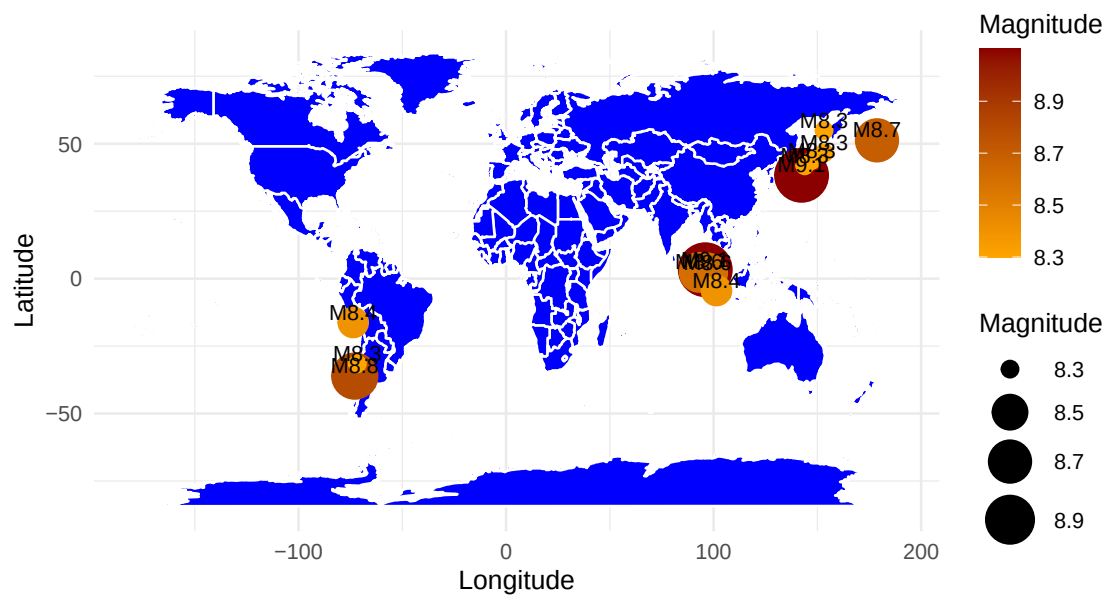


Depth distributions are strongly right-skewed, with most earthquakes occurring at shallow depths. Distinct groupings are also observable, corresponding broadly to shallow, intermediate, and deep-focus earthquakes, which reflect different tectonic processes within the Earth’s lithosphere.

3.2 Largest Earthquakes

Table 2: Top 10 earthquakes by magnitude (1965–2023).

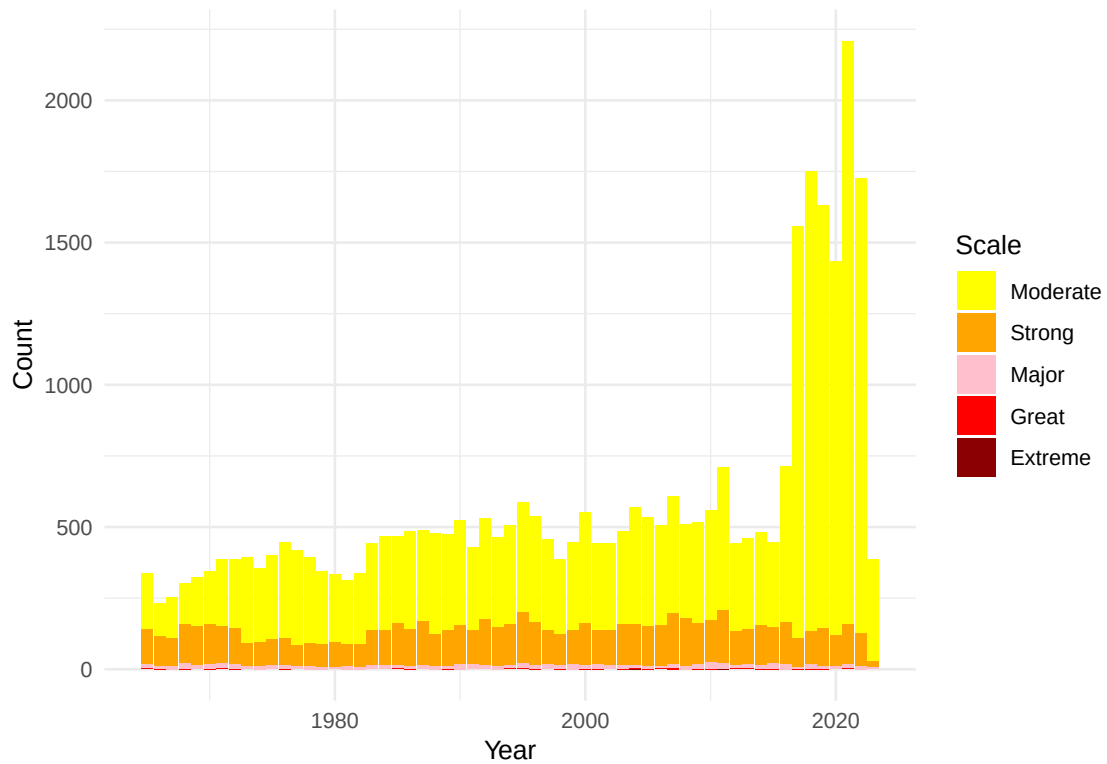
Date	Latitude	Longitude	Depth (km)	Magnitude	Scale
2004-12-26	3.30	95.98	30.0	9.1	Extreme
2011-03-11	38.30	142.37	29.0	9.1	Extreme
2010-02-27	-36.12	-72.90	22.9	8.8	Great
1965-02-04	51.25	178.72	30.3	8.7	Great
2005-03-28	2.08	97.11	30.0	8.6	Great
2012-04-11	2.33	93.06	20.0	8.6	Great
2001-06-23	-16.26	-73.64	33.0	8.4	Great
2007-09-12	-4.44	101.37	34.0	8.4	Great
1994-10-04	43.77	147.32	14.0	8.3	Great
2003-09-25	41.81	143.91	27.0	8.3	Great
2006-11-15	46.59	153.27	10.0	8.3	Great
2013-05-24	54.89	153.22	598.1	8.3	Great
2015-09-16	-31.57	-71.67	22.4	8.3	Great



The ten largest earthquakes in the dataset are predominantly located along major subduction zones, this reinforces the strong relationship between convergent plate boundaries and high-magnitude seismic activity

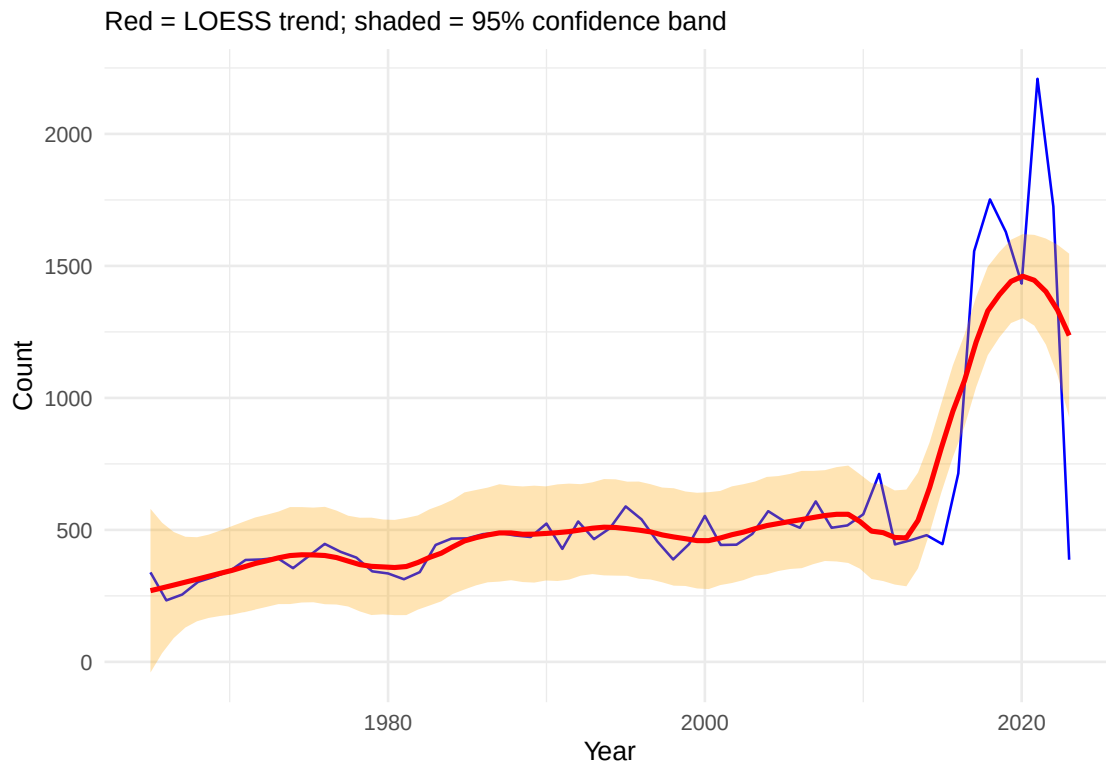
4 Temporal investigation

4.1 Annual Frequency by SScale



Annual earthquake counts vary considerably across magnitude categories. There is a noticeable increase in lower magnitude earthquakes being observed from the late 1990s onwards, which is more likely attributable to improvements in global seismic monitoring than to a true increase in seismic activity. This is important, as it highlights the influence of observational bias in long-term seismic datasets.

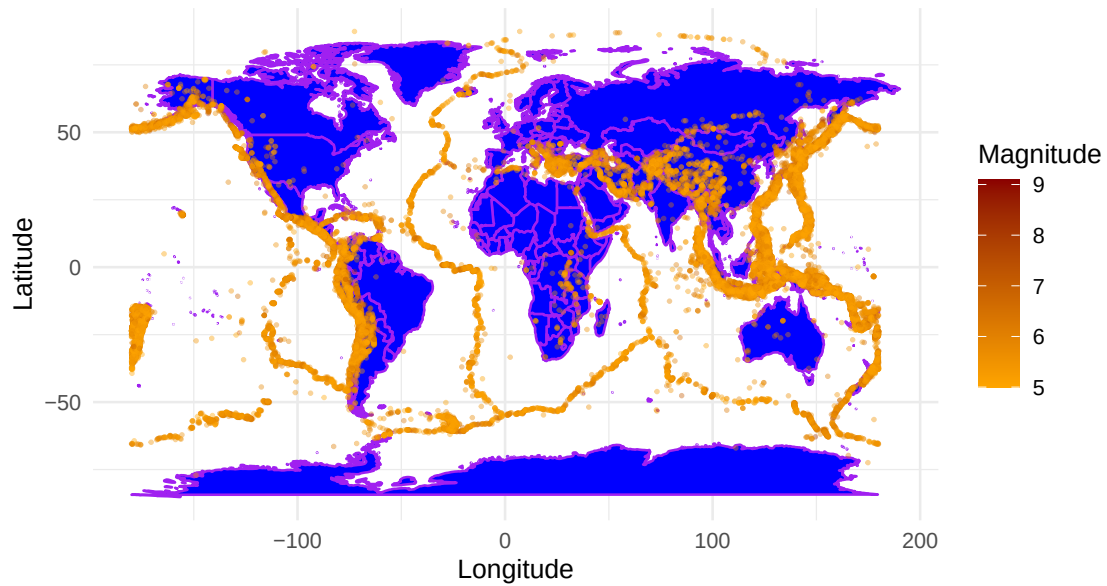
4.2 Magnitude of 5 plus Earthquake trends



When the analysis of earthquakes is restricted to magnitude of 5.0 and above, the long-term trend appears to be relatively stable. Unlike smaller events, these higher-magnitude earthquakes are consistently detectable across the full observation period. This stability suggests that the seen increase in total earthquake counts over time is mostly driven by improvements in detection technology rather than changes in global seismicity. This to me highlights the importance of developing technology in crucial fields like seismology

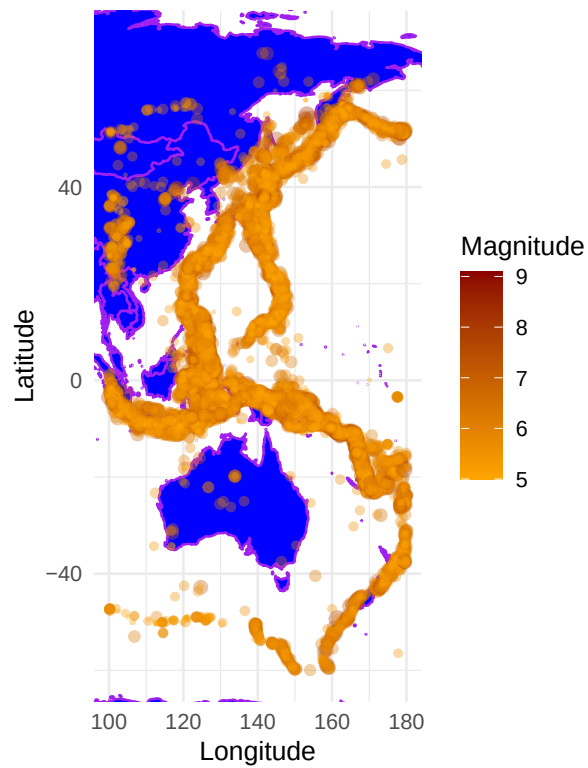
5 Spatial investigation

5.1 Global distribution



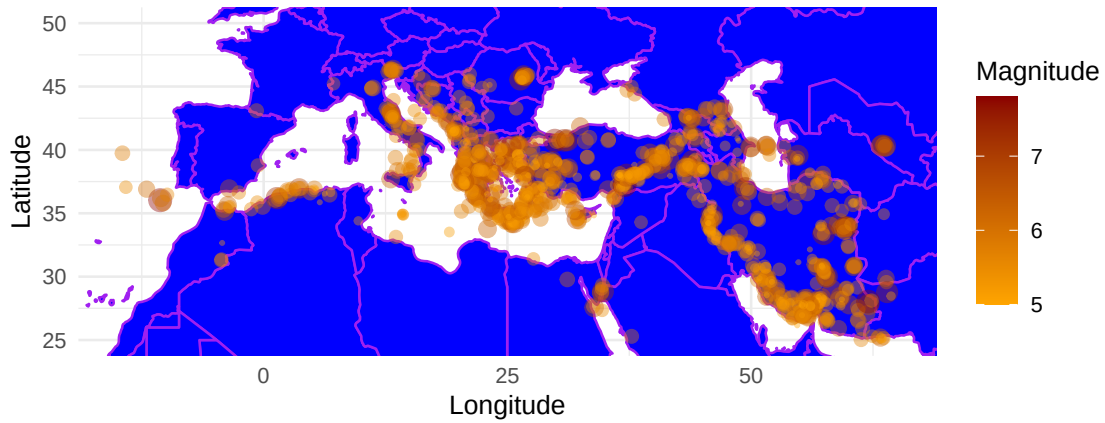
This map shows that global earthquake locations aren't random. They mostly form tight clusters along the edges of tectonic plates. Three main seismic zones stand out: the Circum-Pacific Belt (also called the Ring of Fire), the Alpide Belt stretching from the Mediterranean into Asia, and the system of mid-ocean ridges running across the globe. These patterns show how plate tectonics fundamentally shapes where earthquakes occur around the world.

5.2 Region 1: Western Pacific (Ring of Fire)



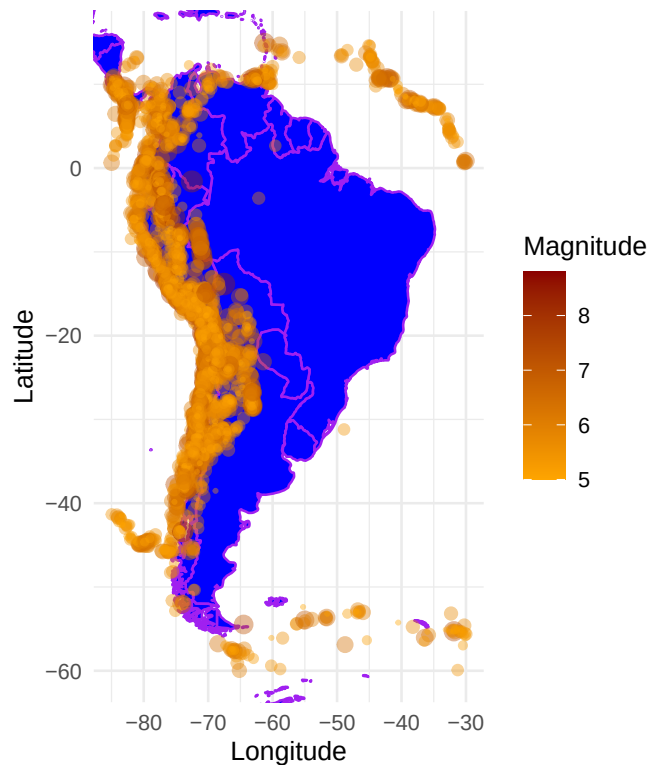
The Western Pacific region shows the highest concentration of earthquakes in the dataset. Countries in areas around Japan, the Philippines, and Papua New Guinea display dense clusters of earthquake activity due to the interaction of multiple tectonic plates. This region experiences frequent earthquakes because of ongoing subduction along the boundaries of the Pacific Plate.

5.3 Region 2: Mediterranean Belt



The Mediterranean region shows high earthquake activity across countries such as Turkey, Greece, Iran, and nearby areas. This is mainly due to the ongoing collision between the African, Arabian, and Eurasian tectonic plates (Jackson & McKenzie, 1984). The pressure created by these plate movements results in frequent moderate to large earthquakes throughout the region.

5.4 Region 3: South American Subduction Zone



In South America, earthquake activity is mainly concentrated along the Andes mountain range, where the Nazca Plate moves beneath the South American Plate (Isacks, Oliver, & Sykes, 1968).

This subduction zone has produced some of the largest earthquakes ever recorded, including the 2010 Chile earthquake, and remains one of the most active seismic regions in the world.

6 Conclusion

This study examined global earthquake patterns using a combined dataset spanning 1965 to 2023. The analysis demonstrated that earthquake frequency decreases with increasing magnitude, consistent with the Gutenberg–Richter relationship.

Temporal trends indicated an apparent increase in smaller earthquakes over time; however, this is likely driven by improvements in global seismic monitoring rather than a true increase in seismic activity. In contrast, higher-magnitude earthquakes exhibit a relatively stable long-term trend.

Spatial analysis confirmed that earthquakes are strongly concentrated along tectonic plate boundaries, particularly within the Pacific Ring of Fire, the Mediterranean–Zagros Belt, and major subduction zones. Overall, the findings reinforce the fundamental relationship between plate tectonics and global seismicity, with large earthquakes occurring primarily in regions of active plate convergence.

7 References

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